

NanoGT Match Report: Biotinidase deficiency

Disease: Biotinidase deficiency (ORPHA:79241)

Primary gene: BTD

Gene CDS: 1632 bp

Inheritance: Autosomal recessive

Target tissues scored: CNS, multisystem

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Vector does not naturally cover all annotated disease tissues: multisystem; Cell-autonomous protein across multiple tissues; high transduction coverage may be required.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Biotinidase enzyme deficiency prevents biotin recycling; liver-expressed BTB deficiency supports systemic gene addition

Gene-addition compatibility: compatible

Preferred modality class: liver gene addition

Evidence level/status: direct / needs_user_fact_check

Evidence summary: BTB LOF is a metabolic enzyme deficiency compatible with liver-directed gene addition

Evidence source: OMIM BTB 253260

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
 - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	OAV101-IT	AAV9	7.4/10	Medium	approved
2	Zolgensma	AAV9	7.4/10	Medium	approved
3	BMN 307	AAV5	7.2/10	Medium	phase2
4	Libmeldy	LV	7.2/10	Medium	approved
5	Strimvelis	LV	7.1/10	Medium	approved

Match #1: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.43	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Biotinidase enzyme deficiency prevents biotin recycling; liver-expressed BTM deficiency supports systemic gene addition
- Gene-addition modality compatibility: supports gene addition

- Mechanism evidence: BTD LOF is a metabolic enzyme deficiency compatible with liver-directed gene addition
- Mechanism source: OMIM BTD 253260 (<https://omim.org/entry/253260>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — BTD standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: multisystem
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway

Dimension	Score	Max	What it measures
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 891bp / cargo 4700bp (19% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Biotinidase enzyme deficiency prevents biotin recycling; liver-expressed BTM deficiency supports systemic gene addition
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BTM LOF is a metabolic enzyme deficiency compatible with liver-directed gene addition
- Mechanism source: OMIM BTM 253260 (<https://omim.org/entry/253260>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

- **ORGANELLE TARGETING: COMPATIBLE** — BTD standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: multisystem
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue

Dimension	Score	Max	What it measures
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.24	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1353bp / cargo 4700bp (29% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Biotinidase enzyme deficiency prevents biotin recycling; liver-expressed BTM deficiency supports systemic gene addition
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BTM LOF is a metabolic enzyme deficiency compatible with liver-directed gene addition
- Mechanism source: OMIM BTM 253260 (<https://omim.org/entry/253260>)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — BTM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: multisystem
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.19	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1521bp / cargo 8000bp (19% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Biotinidase enzyme deficiency prevents biotin recycling; liver-expressed BTM deficiency supports systemic gene addition
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BTM LOF is a metabolic enzyme deficiency compatible with liver-directed gene addition

- Mechanism source: OMIM BTD 253260 (<https://omim.org/entry/253260>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — BTD standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, multisystem
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage

Dimension	Score	Max	What it measures
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.10	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['CNS', 'multisystem'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Disease mechanism: loss of function — Biotinidase enzyme deficiency prevents biotin recycling; liver-expressed BTM deficiency supports systemic gene addition
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BTM LOF is a metabolic enzyme deficiency compatible with liver-directed gene addition
- Mechanism source: OMIM BTM 253260 (<https://omim.org/entry/253260>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — BTM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Gene annotation incomplete; packaging, protein class, and localization scores need manual confirmation